DocMind AI - Project Documentation

Production-Grade AI Documentation Assistant, Vector Search & Code Generator

1. Executive Summary & System Overview

DocMind AI is an advanced, production-grade AI Agent designed to ingest, index, and query complex API documentation across multiple industry formats (PDFs, Markdown, OpenAPI 3.x, and Swagger 2.0 specs). It empowers developers to query APIs using natural English or conversational Hinglish, compare endpoints between different APIs side-by-side, test live HTTP endpoints directly in the browser, and automatically generate production-ready SDK client libraries (Python, JavaScript, Java, cURL) and complete README.md files.

2. Complete Tech Stack & Dependencies

Layer	Technology / Library	Version	Role & Rationale
Backend API	FastAPI	0.109.2	Async REST endpoints & Server-Sent Events (SSE)
ASGI Server	Uvicorn	0.27.1	High-throughput asynchronous web server
Data Schema	Pydantic v2	2.6.1	Strict request validation and data serialization
Vector DB	ChromaDB	0.4.24	Local persistent vector storage with cosine similarity
Embeddings	Sentence-Transformers	2.7.0	Dense 384-dim local embeddings (all-MiniLM-L6-v2)
LLM Inference	Groq Cloud API	0.4.2	Ultra-fast Llama 3.3 70B & 3.1 8B inference
PDF Parser	PyMuPDF (Fitz)	1.23.22	High-speed document parsing & text extraction
Spec Parser	PyYAML / json	6.0.1	Native OpenAPI 3.x & Swagger 2.0 parsing
HTTP Client	HTTPX	0.26.0	Asynchronous HTTP client for live endpoint testing
PDF Export	FPDF2	2.7.9	Dynamic PDF generation for chat history and reports
Frontend UI	HTML5 / CSS3 / ES6+	Native	Clean responsive dark/light UI with voice recognition

3. System Architecture & Processing Pipeline

The DocMind AI architecture consists of three interconnected processing pipelines:
1. Ingestion Pipeline: Ingests uploaded API specs, extracts structured endpoint metadata (methods, paths, parameters, responses), and chunks documents with overlap. Dense 384-dimensional embeddings are generated in batches via all-MiniLM-L6-v2 and stored in ChromaDB.
2. Retrieval & QnA Pipeline: Automatically detects user query language (English vs. Hinglish) and target programming language (Python, JS, Java, cURL). Executes cosine similarity vector retrieval against ChromaDB, builds grounded system prompts, and streams real-time tokens via SSE.
3. Automation & Testing Pipeline: Synthesizes production client SDKs and README.md files on demand, and allows interactive execution of live HTTP requests with custom headers and payload validation.

4. REST API Endpoints Reference

Method & Route	Description	Input Format	Output Format
POST /api/upload	Uploads & indexes API docs	Multipart Form	JSON (chunks stored)
POST /api/detect-name	Auto-detects API name	Multipart Form	JSON (detected name)
GET /api/apis	Lists all indexed APIs	None	JSON (api list)
DELETE /api/apis/{name}	Deletes an indexed API	Path Param	JSON (confirmation)
POST /api/chat	Synchronous grounded QnA	JSON body	JSON (answer + tokens)
POST /api/chat/stream	Real-time streaming QnA	JSON body	Server-Sent Events (SSE)
POST /api/compare	Side-by-side API comparison	JSON body	JSON (dual answers)
POST /api/generate-sdk	Generates client SDK code	JSON body	JSON (code + language)

POST /api/generate-readme	Generates README.md docs	JSON body	JSON (markdown)
POST /api/download-sdk	Downloads SDK source file	JSON body	File Attachment (.py/.js)
POST /api/download-readme	Downloads README.md file	JSON body	File Attachment (.md)
POST /api/test	Executes live HTTP requests	JSON body	JSON (status, time, body)
POST /api/export-pdf	Exports chat to PDF	JSON body	PDF Stream
GET /api/health	System health & statistics	None	JSON (status & counts)

5. Setup, Execution & Verification

To run DocMind AI in a local development or production environment:
1. Environment: Python 3.10+ (Python 3.11 recommended).
2. Virtual Environment: uv venv .venv && source .venv/bin/activate
3. Dependency Installation: uv pip install -r requirements.txt
4. Configuration: Configure GROQ_API_KEY in .env file.
5. Launch Server: uvicorn main:app --host 0.0.0.0 --port 8000 --reload
6. Access Interface: Navigate to http://localhost:8000 in your browser.